

Homework assignment for 02YELMA

Chapter: Electric Fields in Matter

1. An infinitely long dielectric cylinder of radius R is polarized as $\mathbf{P}(r) = \frac{a}{r} \hat{r}$ for $0 < r \leq R$, where a is a constant.
 - (a) Find the bound volume charge density ρ_b .
 - (b) Find the bound surface charge density σ_b at $r = R$.
 - (c) Compute the total bound charge per unit length.
 - (d) Explain why the result is surprising (or not).
 - (e) How can there be surface charge if the divergence is zero everywhere?
2. A dielectric slab occupies the region $-L \leq x \leq L$ and is infinite in the y and z directions. The polarization is given by $\mathbf{P}(x) = kx^2 \hat{x}$ where k is a constant.
 - (a) Find the bound volume charge density $\rho_b(x)$.
 - (b) Find the bound surface charge densities at $x = +L$ and $x = -L$.
 - (c) Compute the total bound charge per unit area.
 - (d) Determine whether the system is overall neutral.
 - (e) Where is charge concentrated more—near the center or near the boundaries? Explain why.
3. A solid sphere of radius R has polarization $\mathbf{P}(r) = kr \hat{r}$.
 - (a) Find ρ_b and σ_b .
 - (b) Calculate the electric field inside and outside the sphere.
 - (c) Write the expression for the potential $V(0)$ at the center of the sphere.
 - (d) Compute the volume contribution and the surface contribution.
 - (e) Add them and simplify.
 - (f) Explain why the two contributions do or do not cancel.